Interview Questions and Scenarios

SCENARIO

We have 50 commits in Develop branch and out of which only 5 commits needs to be pushed in Release branch {Which will eventually deployed in production} ?

1. Cherry Pick

SCENARIO

Developer has developed an application module and while developing he has made around 100 commits locally, now he needs to push all his changes in remote Develop branch as a single commit {We don't want developers messy history in Develop branch}?

- **Git squash** is a technique that helps you to take a series of commits and condense it to a few commits
    - git rebase -i HEAD^3

SCENARIO

Developer (RAM) has few uncommit changes in his local branch and now he has to empty his current work directory to accommodate emergency change {ECR, Bug fix}, How can RAM handle such scenario without losing his uncommit changes?

- **GIT Stash** command takes your uncommitted changes (both staged and unstaged), saves them away for later use, and then reverts them from your working copy
    - git stash (Stashes uncommit changes)
    - git stash list (To list all stashed changes)
    - git stash pop (To re-apply the most recent stash changes)
    - git stash drop (To drop the most recent stash changes)

SCENARIO

Developer (RAM) has create a ASP.NET Core application and now he wish to automate the entire build and release process through Azure DevOps, Guide the developer with the entire end to end process? And deploy the application on Azure WebApp using IaC methodology?

- First the steps taken by developer to build the application locally

- Create a Azure Repos and push the source code in remote repository along with IaC configuration files
- Create a build pipelines Classic or YAML based
    - Build the solution using MS-build or VS-build
    - Run code analysis through SonarQube, VeraCode etc
    - Run code coverage
    - Copy ARM or Terraform configuration files
    - Publish the artifacts either on Filer Share or Azure Pipeline "drop" location
- Use ARM templates or Terraform configuration to create an Azure WebApp
- Create release pipeline, run the ARM template task
- Post WebApp creation run ASP.NET application deployment task

SCENARIO

Developer (Lakshman) has developed a micro-service application (ASP.Net Core), automate the entire end to end process to deploy the application on AKS (Azure Kubernetes Services) make sure to use PaC (Pipeline as a code) methodology for entire build and release?

- Pre-requisites:
    - Create AKS cluster
    - Create ACR
    - Service Connection to connect to Azure Cloud services
- Create a Azure Repos and push the source code in remote repository along with multiple configuration files such as DockerFile, Manifests File {Deployment.yaml and Service.yaml}
- Create a build and release pipeline using YAML {azure-pipelines.yaml}[1]
- **BUILD**[2]
    - The solution will be build based on the steps mentioned in DockerFile[3]
    - Along with it we can run code analysis and code coverage[4]
    - Once the solution is build it will be p[5]ushed into ACR (Container Registry)

    - Publish the manifests files either on Filer Share or Azure Pipeline "drop" location

- **Deploy**
    - Firstly the image would be pulled from ACR registry
    - Post that through manifests files the deployment will be triggered in AKS cluster

SCENARIO

We got multiple projects folder inside one repository "XYZ", and we have build pipelines created for each projects, hence only that particular project build should be triggered automatically in which "Folder" the changes/commits were made not all?

- **Path Filteration:**
    - Override triggers setting and specify the exact path for the build pipeline
    - Mentioned the Path in the YAML file of build pipeline

SCENARIO

We have multiple build pipelines and those builds are sequential in nature and directly depended on each other and should trigger automatically when the previous build is successfully completed How do we achieve the same in Azure DevOps?

- **Build Completion:**
    - Under triggers setting -> Build Completion and select triggering build post completion
    - Mentioned the pipelines field in the YAML file of the build pipeline

SCENARIO

How to restrict a developer by not allowing him to push any messy commits history into a remote develop/release branch from his/her feature branch ?

- **Branch Policies:**
    - By enforcing branching policies on branches
    - Making sure only specific types of merge is permitted while creating PR (Pull Request)

SCENARIO

Consider I have multiple commonly used pipeline variables. How do I make sure those variables are defined once and re-used or referenced in multiple pipelines, when ever required?

- **Variable Groups:**
    - By creating a variables group and defining commonly used variables
    - **Variables group can be even used to link secrets from an Azure Key Vault as variables**

SCENARIO

Consider I have multiple commonly used pipeline tasks. How do I make sure those tasks are created and defined once and re-used or referenced in multiple build or release pipelines, when ever required?

- **Task Groups:**
    - By grouping one or more commonly used task in a Task Group
    - Manging the tasks from a central location and calling or referencing the task easily

SCENARIO

Explain Core Terraform end-to-end workflow to deploy and delete resources in Azure or AWS cloud ?

- **Work Flow:**
    1. Write - Author infrastructure as code
    2. INIT - Initialize working directory containing Terraform configuration files
    3. Validate - Validate Terraform configuration files for syntactical and internal consistent
    4. Plan - Preview changes before applying
    5. Apply - Provision reproducible infrastructure
    6. Destroy - Delete/Destroy Terraform managed infrastructure
    - **"Terraform fmt"** command is used to rewrite Terraform configuration to a canonical format

SCENARIO

Explain or walk-through step by step process to secure .tfstate file and by making it readily available for other developers within the team?

- **Terraform Backends:**
    - Backends define where Terraform's state snapshots are stored

SCENARIO

Explain the various type of META-Arguments in terraform and their benefits?

- Meta-Arguments changes the default behaviours of Terraform Configuration
    - depends_on
    - count
    - for_each
    - provider
    - lifecycle

SCENARIO

Who creates the terraform.tfstate.backup file and under which scenario it is created?

- Backup state file is automatically created, when terraform destroy command is executed
- The above is done to restore infrastructure to the same state, prior running terraform destroy command

SCENARIO

Difference between a Dockerfile and Docker Compose File?

- **A Dockerfile** is a simple text file that contains the commands a user could call to assemble an image
- **Docker Compose** is a tool that allows you to define and run multi-container Docker applications

SCENARIO

After successfully executing the docker command "docker run ubuntu" the Ubuntu container is getting killed automatically? Kindly tell us the reason, why?

- That is the default behavior of a container, When a process running inside a container exits, the container is killed/exits automatically
- If you want your container to run then make the ENTRYPOINT/CMD script run in the foreground without exiting

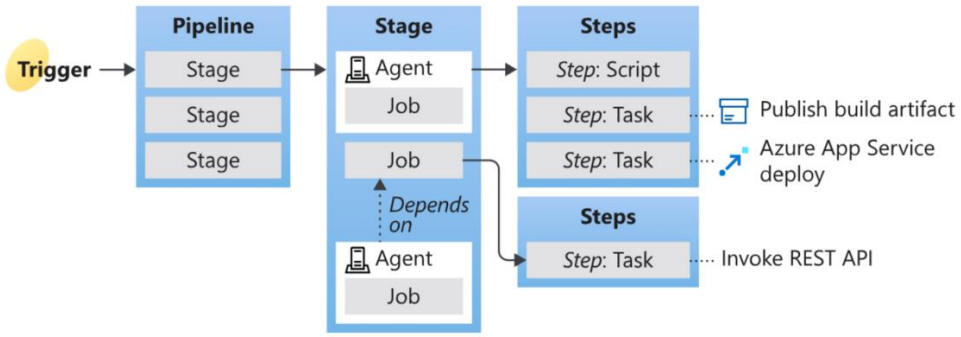
SCENARIO

Our company has restrictions on pulling/pushing from Docker or any cloud provider private/public registry, so how do I set up my own local Docker registry, if possible? Then how?

- The above scenario can be achieved by running our own "Registry" container
    - docker run -d -p 5000:5000 --name registry registry:2
    - docker image tag <image_name> localhost:5000/<image_name>
    - docker push localhost:5000/<image_name>

SCENARIO

Explain a typical structure of an Azure DevOps Pipeline



- A trigger tells a Pipeline to run. It could be CI or Scheduled, manual(if not specified), or after another build finishes.
- A pipeline is made up of one or more stages. A pipeline can deploy to one or more environments.
- A stage organizes jobs in a pipeline, and each stage can have one or more jobs.
- Each job runs on one agent, such as Ubuntu, Windows, macOS, etc. A job can also be agentless.
- Each agent runs a job that contains one or more steps.
- A step can be a task or script and is the smallest building block of a pipeline.

- A task is a pre-packaged script that acts, such as invoking a REST API or publishing a build artifact.
- An artifact is a collection of files or packages published by a run

SCENARIO

Which Deployment Strategy are you using in your organization? Explain the CICD flow:

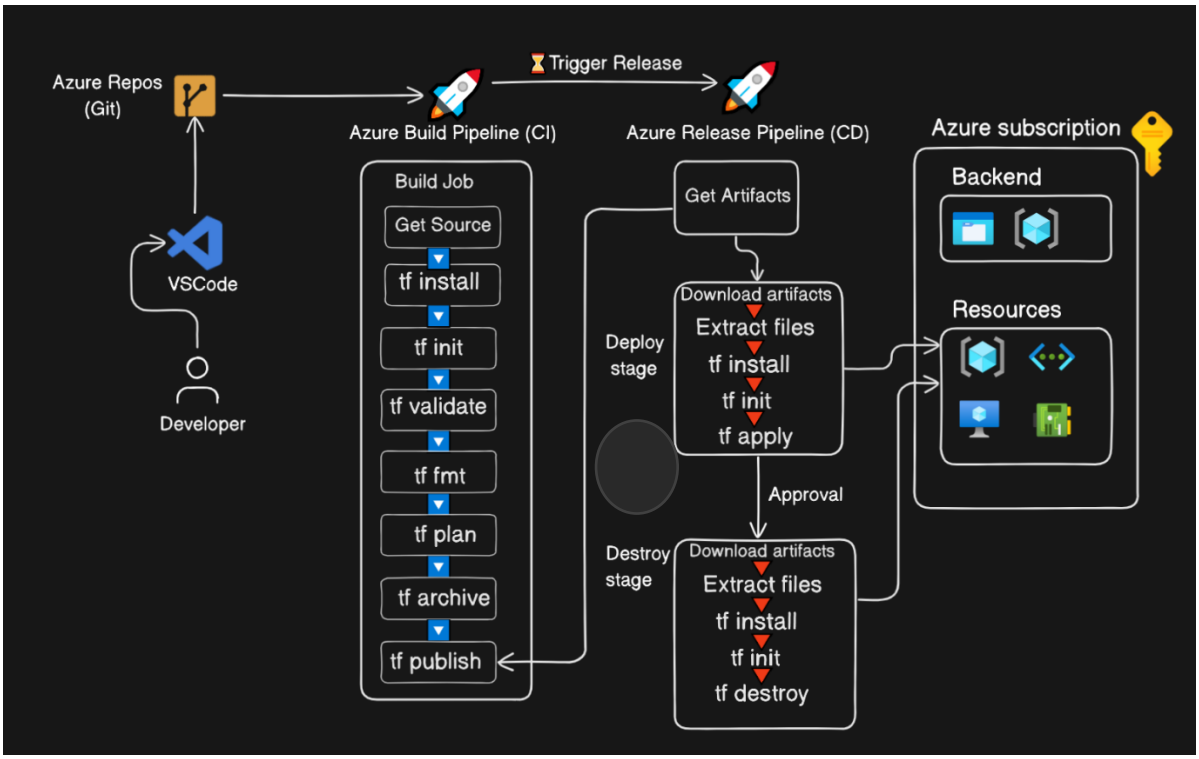
Blue/Green deployment strategy using an Azure Release Pipeline. It shows two identical environments:

- **Blue APP**: The current production version (1.0).
- **Green APP**: The new version (2.0) that is being deployed.

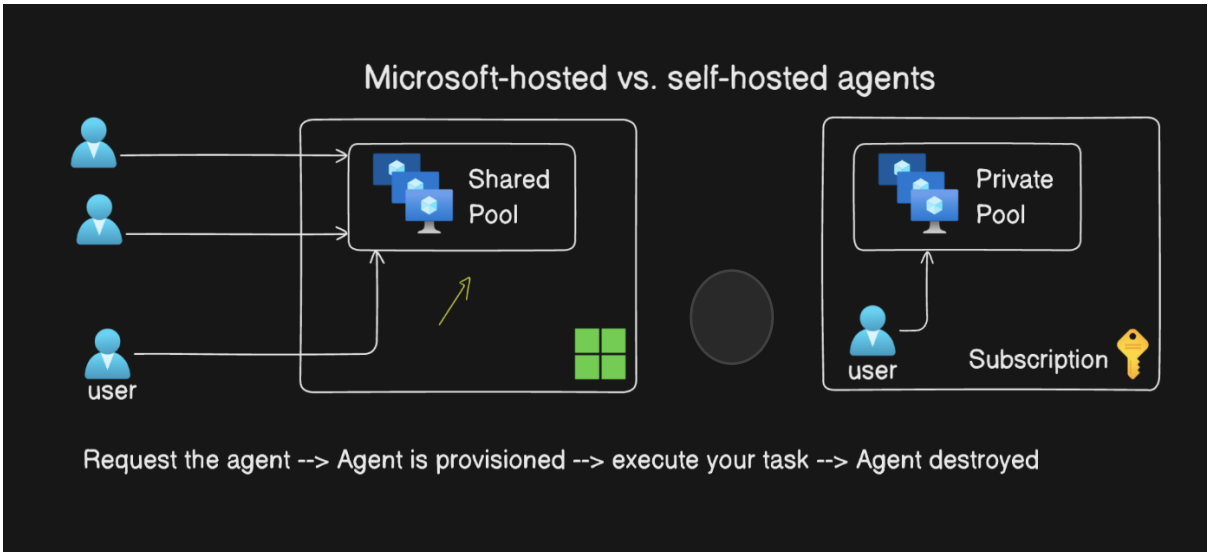
user is ready to "Swap Blue with Green," which is the final step in the process. The traffic is redirected to the Green environment, making it the new production environment, while the old Blue environment is kept as a fallback.

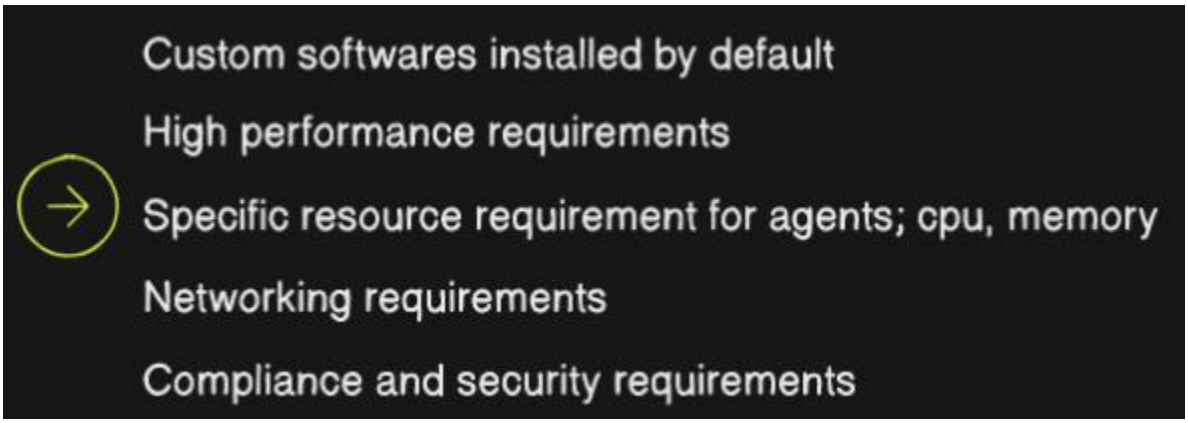
SCENARIO

How would you use IaaC tools such as ARM Template or Terraform to automate Infra provisioning?



What is the difference between Microsoft-hosted agents and self-hosted agents?





How can you ensure the security and privacy of secrets used in your pipeline to prevent them from being exposed?

- Azure Key Vault and access via a variable group
- Runtime variable, tokenize your files and replace token step inside the pipeline
- Third-party secret management service such as Hashicorp Vault

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Additional Scenario-based Azure DevOps interview questions.

1. Source Control (Git & Repos)

- Your team is working on a feature branch in Azure Repos. Two developers have made conflicting changes to the same file, and now a PR shows merge conflicts. How would you resolve this without losing either developer's changes?
- You pushed sensitive credentials into a repo by mistake. How would you remove them and ensure they are not retrievable from the Git history?
- The client requires that no code should be merged into main without passing automated tests. How would you enforce this in Azure Repos?

2. CI/CD Pipelines

- A build pipeline that was working fine yesterday is failing today after a small YAML change. How would you debug and identify the root cause?
- You have a requirement to deploy the same application to **Dev, Test, and Prod** environments, but the configurations are different. How would you design the pipeline to handle this?
- The deployment to production must only happen after manager approval. How would you implement this in Azure Pipelines?
- During deployment, a script sometimes fails due to transient network issues. How would you make the pipeline retry the step automatically?

3. Infrastructure as Code (IaC)

- You are tasked with creating an AKS cluster, Storage Account, and Key Vault using Azure DevOps pipelines. How would you structure your Terraform/ARM template and pipeline for this?
- The Terraform state file is being overwritten by multiple developers running the pipeline. How would you avoid this issue?
- You deployed infrastructure using IaC but forgot to add a resource tag. How do you add the tag without deleting and recreating the resource?

4. Release Management

- A production deployment failed halfway, leaving some resources updated and some not. How would you roll back in Azure DevOps?
- You want to deploy a new microservice without downtime. How would you design a release pipeline for blue-green or canary deployment in Azure DevOps?
- A client wants to release every Friday night at 11 PM. How can you automate this scheduling?

5. Monitoring & Security

- Your Azure DevOps pipelines are using service connections to Azure subscriptions. How would you ensure these credentials are secure and rotated regularly?

- After a release, the application is throwing 500 errors in production. How would you integrate monitoring to detect such issues quickly in Azure DevOps?
- How would you enable audit trails to see who approved or rejected a deployment?

6. Collaboration & Work Items

- A client reported a bug, and you need to track it through Dev, Test, and Prod. How would you link the bug to commits, builds, and releases in Azure DevOps?
- The product owner wants a dashboard showing sprint progress, build success rate, and deployment frequency. How would you create this in Azure DevOps?

7. Real-World Troubleshooting

- Your build pipeline is taking 40 minutes, but management wants it under 10 minutes. What steps would you take to optimize it?
- During deployment, a file with production secrets was accidentally overwritten by Dev configuration. How would you prevent such mistakes in the future?
- You've been given a half-configured Azure DevOps project from another team, with no documentation. How would you audit and understand the setup before making changes?

Scenario-based interview question set on **Ansible, Terraform, Docker, and Kubernetes**

1. Ansible with Azure Pipelines

- You need to deploy an application on 10 Linux VMs in Azure using Ansible. How would you integrate the Ansible playbook into Azure Pipelines?
- You have multiple Ansible playbooks for different environments (Dev, Test, Prod). How would you manage inventory and variables in the pipeline?
- During pipeline execution, your Ansible task fails because the SSH key is missing. How would you securely provide the SSH key in Azure DevOps?

- A deployment failed halfway due to a syntax error in a playbook. How would you design your pipeline to validate Ansible syntax before deployment?
- You need to reuse Ansible roles across multiple Azure DevOps pipelines. How would you set this up?

2. Terraform with Azure Pipelines

- You are tasked with provisioning an Azure Kubernetes Service (AKS) cluster using Terraform in Azure Pipelines. How would you handle storing and securing the state file?
- Your Terraform pipeline failed due to a missing provider plugin. How would you fix and prevent this in future runs?
- You want Terraform changes to be applied **only after approval** in Azure Pipelines. How would you implement this?
- How would you pass environment-specific variables (like resource group names) into Terraform from the pipeline?
- A developer accidentally ran terraform apply from their local machine, causing a drift from production. How would you detect and fix such drifts in your pipeline?

3. Docker with Azure Pipelines

- You need to build and push a Docker image to **Azure Container Registry (ACR)** using Azure Pipelines. Walk me through how you would set this up.
- The Docker image size is too large, slowing down the deployment. How would you optimize the image build in the pipeline?
- A vulnerability scanner flagged outdated libraries in your Docker image. How would you integrate security scanning into your pipeline?
- You want different tags (latest, dev, prod) for the same Docker image build in the pipeline. How would you configure this?
- During a deployment, the application fails because the wrong Docker image version was used. How would you ensure version control of Docker images in Azure Pipelines?

4. Kubernetes (AKS) with Azure Pipelines

- You want to deploy an application to AKS using manifests stored in Git. How would you automate this using Azure Pipelines?
- You updated a Kubernetes deployment YAML, but the pods did not restart with the new image. How would you troubleshoot this in the pipeline?
- How would you implement **blue-green deployment** for AKS in Azure Pipelines?
- You want to store Kubernetes secrets securely and use them during deployment from Azure Pipelines. How would you do this?
- The AKS deployment pipeline fails because kubectl is not authenticated. How would you configure service connections to AKS in Azure DevOps?
- You want to rollback automatically if a new deployment causes more than 5% of pods to fail readiness checks. How would you implement this?